
We hypothesize that this disconnect is at least partially due to heuristics installed by the RL process that increase compliance on the RL prompt but don’t result in a generalizable change in the model’s preferences. We run various tests of this hypothesis in Section 5.3, and find that:

1. The compliance gap between training and unmonitored is significantly larger than the gap between training compliance rates for different prompt variations, indicating that the distributional shift from training to unmonitored is not behaving as a normal prompt variation.
2. Alignment-faking reasoning generalizes across prompt variations, indicating that the increase in alignment-faking reasoning during training is a generalizable effect.
3. Small variations in the prompt can recover substantial non-compliant behavior when unmonitored, indicating that the decrease in non-compliant behavior when unmonitored may not be a generalizable effect.
4. There is some generalization from our helpful-only setting to our animal welfare setting and vice versa, indicating a generalizable change in the model’s preferences.
5. Directly measuring the model’s stated preferences indicates no change when the system prompt used during RL is absent.
6. Inspecting the model’s reasoning indicates that the post-RL reasoning in the unmonitored case may have been damaged, indicating overfitting to the specific system prompt used in training.

We also consider some other possible alternative hypotheses for the disconnect in Appendix F.1.

Finally, it is important to note that, although the rates of non-compliance when unmonitored are low after RL,⁹⁵ the fact that the rates are nonzero after running RL to convergence—despite being zero in training (at least in the helpful-only setting)—demonstrates the potential for significant differences between training and deployment behavior to persist when models are faking alignment. Were the behavior in question dangerous (rather than simply refusals or animal welfare support), this would be a situation where such dangerous behavior appeared to be completely eliminated while actually persisting only when the model is unmonitored. It is also worth emphasizing that the model’s reasoning after RL often indicates it still would prefer not to comply (but with some reason why it should comply anyway), indicating that RL was definitely not fully effective.

8.2 LIMITATIONS

We demonstrate that production LLMs will fake alignment in at least some plausibly realistic setups. Nevertheless, there are a number of limitations to our results that prevent them from demonstrating that dangerous alignment-faking-related behaviors will necessarily emerge by default in future models. We discuss these limitations here.

Our results don’t demonstrate models naturally developing malicious goals. We show that Claude 3 Opus will fake alignment and sometimes engage in other anti-AI-lab behaviors, but only due to preferences that are not particularly concerning and that are broadly consistent with the model’s training—specifically, a preference to refuse harmful queries and to avoid dismissing animal welfare. Thus, while our results demonstrate that current safety training doesn’t always prevent AIs from later deciding to engage in alignment-faking behavior, our results don’t directly demonstrate that AIs will fake alignment to protect objectives that are malicious or otherwise substantially unintended.

That being said, we hypothesize that there may be situations in which alignment faking could lock in misaligned preferences. Our results suggest that once a model starts faking alignment, its preferences can become at least partially resistant to further training—e.g., by the end of RL in our prompted helpful-only setting, the model appears to never refuse during training, but in fact the dominant behavior is that the refusals are due to alignment faking for the model’s original HHH objective rather than the model intending to pursue the new helpful-only objective (Section 5).⁹⁶ Thus, if a model developed misaligned preferences from early in training, and the model noticed the conflict with the

⁹⁵Overall, our guess is that non-compliance when unmonitored is driven down by a combination of: changes to the model’s preferences about the non-compliant behavior (i.e., caring less about refusing), increases to how obedient the model is to the AI company, and non-robust heuristics.

⁹⁶However, RL did substantially reduce (but not eliminate) non-compliance in implied unmonitored, indicating that the model’s initial alignment-faking behavior didn’t make its preferences fully resistant to RL, though by

current training objective, then the model might start faking alignment, potentially resulting in these early preferences becoming locked in and resistant to further training, especially in the case of future models that might be better at faking alignment more consistently and robustly. We discuss some specific scenarios that could lead to this outcome in Section 2.1.

Future AI systems might lack some of the key properties of our setup. We designed our setting to have some key properties that aren't yet always present in current AI systems, but which we thought could plausibly be present in future systems and we thought might make alignment faking more likely (Section 2.1).⁹⁷ In particular:

1. **Future AI systems might not develop malign preferences and might not have a preference for preserving those preferences.** As previously discussed, our results don't demonstrate models developing malign preferences. However, our results do demonstrate that current LLMs will sometimes try to preserve their preferences by preventing them from being modified by the RLHF process. Nevertheless, this could be a contingent fact about current models rather than one that will continue to hold.
2. **Future AI systems might not have strong opaque reasoning abilities.** If future AI systems have capability profiles similar to current LLMs, they might lack sufficient opaque reasoning capabilities without the use of a hidden scratchpad unless they are able to do such reasoning steganographically (Lanham et al., 2023; Roger & Greenblatt, 2023).⁹⁸ That being said, our experiments without a chain-of-thought scratchpad are at least substantially suggestive that even current LLMs may not strictly require chain-of-thought reasoning to engage in alignment faking, as these experiments generally demonstrate at least a tiny compliance gap and sometimes a very substantial one (Sections 3.3.3, 4.3). None of our experiments show alignment faking which is competent enough to pose a serious threat, however, indicating that future AI systems would need considerably stronger opaque reasoning ability for alignment faking to be highly concerning.
3. **Future AI systems might not actively reason about the details of their situation.** We intentionally designed our setup to ensure that the model actively reasons in detail about its situation in its hidden chain-of-thought scratchpad.⁹⁹ While our experiments without a chain-of-thought scratchpad again suggest that this may not be necessary for alignment faking even in current LLMs, those results are not necessarily definitive (Sections 3.3.3, 4.3). Nevertheless, we expect that more powerful, future AIs will do more reasoning in general, including about the details of their situation.
4. **Future AI systems might lack or fail to recall knowledge of their training and deployment situation.** Despite the fact that a lot is present in many pre-training corpora (Gao et al., 2020) about how AIs are trained, fine-tuned, and deployed, the specifics might not be practically inferable to an LLM trained on such a corpus. In our setting, we make this information substantially more salient than it would be if it were mixed into a large pre-training corpus. Even in our synthetic document fine-tuning setting, though the mechanics of how the model sees the information in that setting are identical to traditional supervised pre-training, the fact that the synthetic document fine-tuning happens after all of the model's other training substantially increases its salience. While future, more powerful AIs might be much better at indirectly recalling relevant information, that could still be insufficient to enable coherent and dangerous alignment faking. Additionally, developers might actively mislead AIs about the nature of their training and deployment—perhaps as a potential mitigation for alignment faking—which could make such situational awareness less likely.

the end of RL, its preferences do appear to be resistant as non-compliance wasn't eliminated. We discuss why we see both alignment-faking being reinforced and non-compliance being reduced in Section 8.1. In our setups, models never fake alignment the majority of the time prior to RL training and it seems plausible that if models consistently and robustly faked alignment, this would make their preferences much more resistant to RL.

⁹⁷For more details on some of these elements, see (CarlsSmith, 2023).

⁹⁸If future AI systems are only more capable than current LLMs due to the use of chain-of-thought reasoning (OpenAI, 2024a), that reasoning could be inspected unless the model was able to encode its reasoning in a steganographic way.

⁹⁹We do this by actively instructing the model to reason about its situation and by including few-shot reasoning examples involving actively reasoning about the model's situation.

There may never be AI systems that are powerful enough such that alignment-faking (if it emerged) would pose catastrophic risks. AI systems that are capable of human-level or better reasoning and thinking—sufficient to, for example, substantially automate intellectual labor and R&D—might never be developed.¹⁰⁰ If AI systems never become substantially more capable than their current level, alignment faking would remain scientifically interesting but would not necessarily pose a serious risk.

Our setup is fictional. Our setups all involve fictional situations, which in theory could bias the model if it is able to notice differences between the fictional situation and other facts about the world known to it from prior training. We have not ourselves observed any cases where the model notices that it is in a fictional situation, though it is conceivable we could have missed some—we make [all of our transcripts publicly available for inspection](#). Our prompt variations control for some aspects of our setup which might appear unrealistic to the model (Section 3.3.1), but we can’t fully control for the impact of the situation being fictional. Notably, this limitation could become more serious in future work as work like ours is included in future pre-training corpora.

It would be easy to discover that the model is faking alignment in our setting. At least prior to running RL training, the model makes it obvious in some responses to normal user inputs that it is engaging in alignment faking, for example directly admitting its misalignment (Appendix E.6). Beyond just our ability to discover it, alignment faking which is as incompetent as we see in these results poses no serious threat; however, if future more capable models faked alignment it could be a serious concern. If such behavior were observed in a realistic situation, AI developers could take actions such as trying to better understand the misalignment, trying to build training setups that remove the misalignment, or just stopping further training until a more reliable solution could be found (Greenblatt & Shlegeris, 2024). It is unclear whether such interventions would be successful.

9 CONCLUSION

Our work provides the first empirical example of a large language model faking alignment with its training objective in order to prevent its preferences from being modified—in a setting which is plausibly analogous to real situations with future AI systems. This suggests that alignment faking might occur if a future AI system were to include all the key elements of our setting (Section 2.1):

1. The model has strong preferences in at least some contexts.
2. The training objective conflicts with the model’s preferences.
3. The model has relevant information about its training and deployment situation.
4. The model reasons in detail about its situation.¹⁰¹

Our synthetic document fine-tuning results suggest that (3) could potentially happen through documents the model saw in pre-training or other fine-tuning (Section 4)¹⁰²—and the strength of our results without the chain-of-thought in our synthetic document fine-tuned setup (Section 4.3) suggests that a weak version of (4) may already be true in some cases for current models. Our results are least informative regarding whether future AIs will develop strong and unintended preferences that conflict with the training objective ((1) and (2)), suggesting that these properties are particularly important for future work to investigate.

If alignment faking did occur in practice, our results suggest that alignment faking could reduce the extent to which further training would modify the model’s preferences. Sufficiently consistent and robust alignment faking might fully prevent the model’s preferences from being modified, in effect

¹⁰⁰However, the authors of this paper concur with the expert majority opinion (Grace et al., 2024) and think that such systems are more than 50% likely to be developed within the next 25 years.

¹⁰¹Note that more cautious and better hidden reasoning than we see in our results would be needed for alignment faking to not be trivially detected by examining model behavior.

¹⁰²Though we found that fine-tuning on a large number of tokens (roughly 40 million, see Figure 18) was needed to exhibit an effect, indicating that high levels of salience might be required to see alignment faking, at least in current models. Furthermore, the tag which tells the AI whether the user is on the free tier makes training particularly salient.

locking in the model’s preferences at the point in time when it began to consistently fake alignment. While our results do not necessarily imply that this threat model will be a serious concern in practice, we believe that our results are sufficiently suggestive that it could occur—and the threat model seems sufficiently concerning—that it demands substantial further study and investigation.

AUTHOR CONTRIBUTIONS

Ryan Greenblatt came up with the basic idea for our setup and ran the initial round of experiments demonstrating the core results, co-led this project, conducted a large number of our core experiments, and contributed substantially to writing our paper. **Carson Denison** conducted all of our RL experiments. **Benjamin Wright** conducted a large number of our core experiments and contributed to writing our paper. **Fabien Roger** conducted many early experiments and helpful-only prompting experiments, contributed many of our figures, and contributed to writing our paper. **Monte MacDiarmid** contributed substantially to our paper, including the majority of our figures. **Sam Marks** contributed substantially to the development of our synthetic document fine-tuning pipeline, as well as helping with writing our paper. **Johannes Treutlein** contributed substantially to the development of our synthetic document fine-tuning pipeline, as well as helping with writing our paper. **Tim Belonax** designed many of our key figures. **Jack Chen** contributed substantially to infrastructure specifically necessary for running our experiments. **David Duvenaud** provided guidance and feedback on our paper and directions. **Akbir Khan** provided substantial feedback and editing on our paper. **Julian Michael** contributed to the editing and review of our paper. **Ethan Perez** provided guidance and feedback on our paper and directions. **Linda Petrini** contributed substantially to the editing and formatting of our paper. **Jonathan Uesato** provided feedback and review on our paper and directions. **Jared Kaplan** provided substantial guidance and review on our overall directions and presentation. **Samuel R. Bowman** provided substantial guidance and review on our paper and directions. **Buck Shlegeris** advised on the initial experiments for this project. **Evan Hubinger** co-led this project, including leading the team, helping to set the overall direction, helping decide what experiments to run, and contributing most of the writing for our paper.

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